# Isaac ROS 3D Scene Reconstruction and Mapping

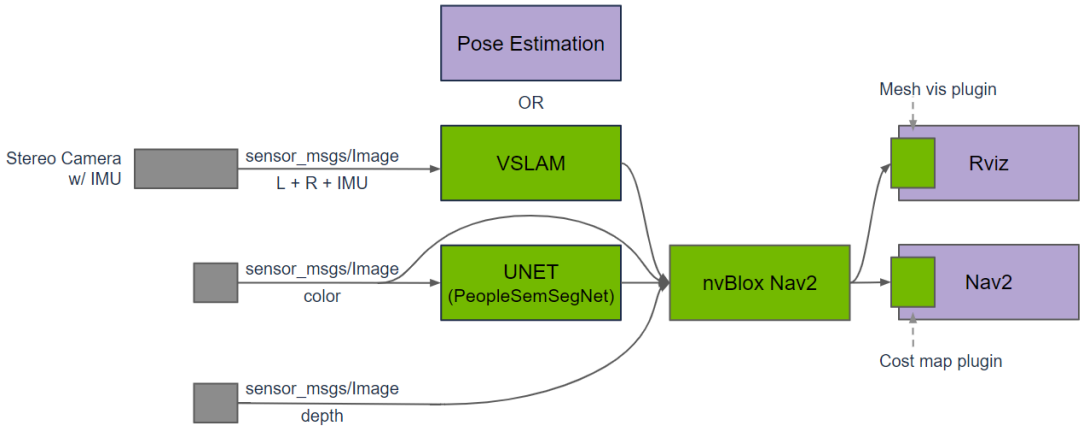

**Isaac ROS 3D Scene Reconstruction and Mapping**

- Overview
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Isaac ROS Nvblox official website link: [https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_nvblox/index.html](https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_nvblox/index.html)

## Overview

Nvblox uses depth and pose data to reconstruct a 3D scene in real time, and can output a 2D costmap for Nav2. It can work with a depth camera or a 3D LiDAR to provide an environment representation for robot navigation.




## Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`; each terminal should use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
     STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

**Note: The terminals below all execute inside the container:**

Terminal A: Start Nvblox:

```
ros2 run nvblox_ros nvblox_node --ros-args \
  --params-file
/opt/ros/jazzy/share/nvblox_examples_bringup/config/nvblox/nvblox_base.yaml \
  --params-file
/opt/ros/jazzy/share/nvblox_examples_bringup/config/nvblox/specializations/nvblox
_sim.yaml \
  -p use_sim_time:=false \
  -p global_frame:=odom \
  -p map_clearing_radius_m:=-1.0 \
  -p layer_visualization_exclusion_radius_m:=0.0 \
  -p decay_tsdf_rate_hz:=0.0 \
  -p num_cameras:=1 \
  -r camera_0/depth/image:=/front_stereo_camera/depth/ground_truth \
  -r camera_0/depth/camera_info:=/front_stereo_camera/left/camera_info \
  -r camera_0/color/image:=/front_stereo_camera/left/image_raw \
  -r camera_0/color/camera_info:=/front_stereo_camera/left/camera_info
```

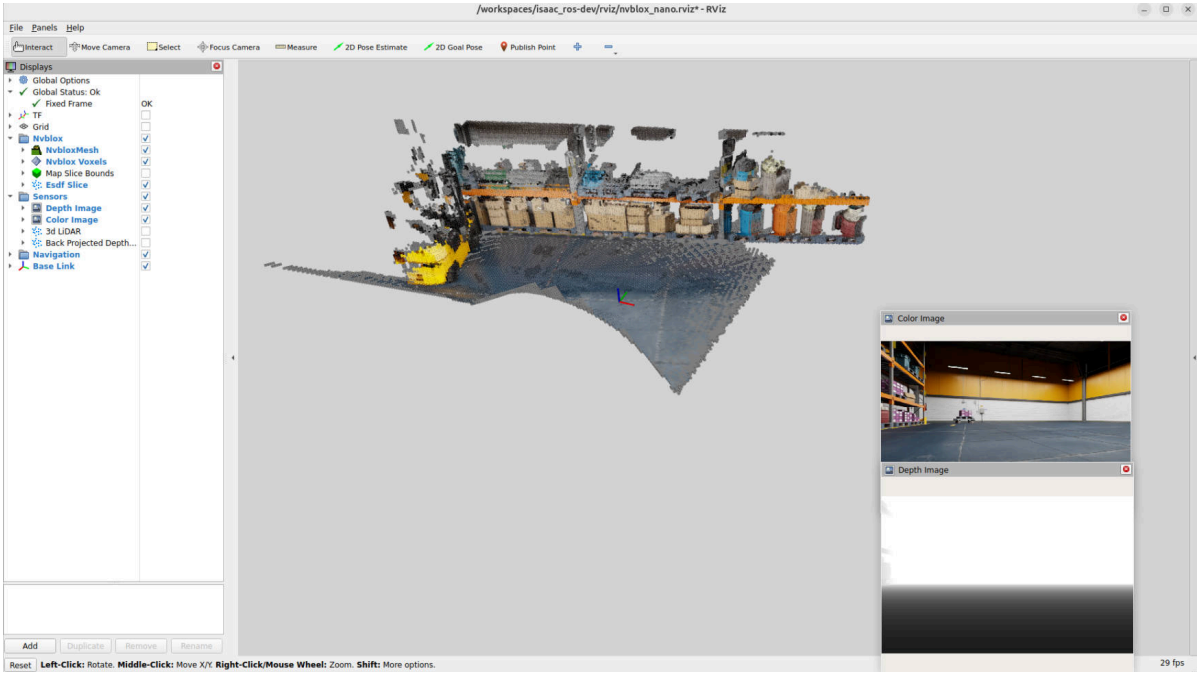
Terminal B: Open the dedicated RViz configuration:

```
rviz2 -d /workspaces/isaac_ros-dev/rviz/nvblox_nano.rviz
```

The configuration displays `/nvblox_node/mesh`, `/nvblox_node/color_layer`, and ESDF.

Terminal C: Play the quickstart for a single round only:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_nvblox/quickstart
```




After playback ends, the bag exits by itself; Nvblox and RViz keep running, the image stops at the last frame, and the map in memory continues to be displayed. Another terminal can be opened to check:

```
ros2 topic echo --once /nvblox_node/static_map_slice
```